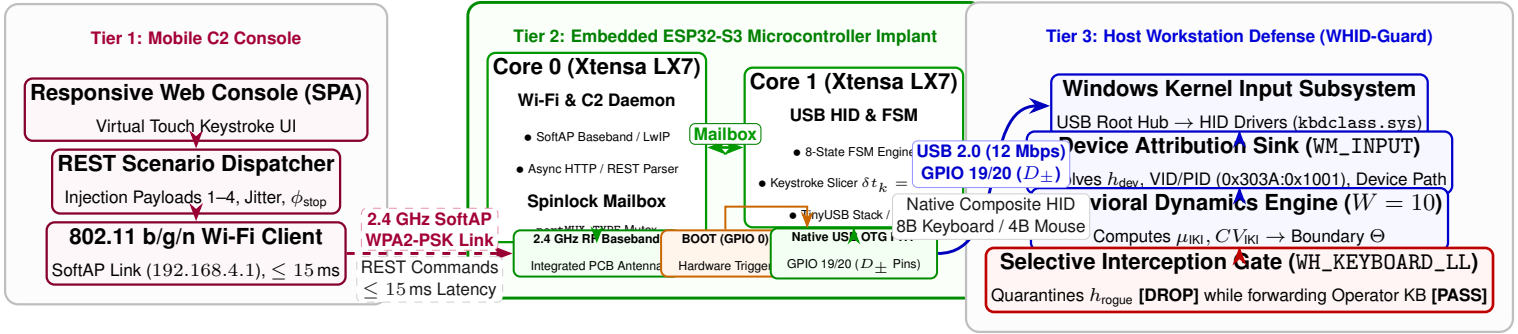

Tier 1: Mobile C2 Console
Responsive Web Console (SPA)
Virtual Touch Keystroke UI
REST Scenario Dispatcher
Injection Payloads 1–4, Jitter, ϕ_{stop}
802.11 b/g/n Wi-Fi Client
SoftAP Link (192.168.4.1), ≤ 15 ms
2.4 GHz SoftAP
WPA2-PSK Link
REST Commands
≤ 15 ms Latency
Tier 2: Embedded ESP32-S3 Microcontroller Implant
Core 0 (Xtensa LX7)
Wi-Fi & C2 Daemon
SoftAP Baseband / LwIP
Async HTTP / REST Parser
Spinlock Mailbox
Mailbox
Core 1 (Xtensa LX7)
USB HID & FSM
8-State FSM Engine
Keystroke Slicer $\delta t_k =$
TinyUSB Stack /
2.4 GHz RF Baseband
Integrated PCB Antenna
BOOT (GPIO 0)
Hardware Trigger
Native USB OTG
GPIO 19/20 ($D_{\pm}$ Pins)
USB 2.0 (12 Mbps)
GPIO 19/20 ($D_{\pm}$)
Native Composite HID
8B Keyboard / 4B Mouse
Tier 3: Host Workstation Defense (WHID-Guard)
Windows Kernel Input Subsystem
USB Root Hub → HID Drivers (kbdclass.sys)
Device Attribution Sink (WM_INPUT)
olves h_{dev}, VID/PID (0x303A:0x1001), Device Path
vioral Dynamics Engine ($W = 10$)
Computes $\mu_{\text{IKI}}, CV_{\text{IKI}}$ → Boundary Θ
Selective Interception Gate (WH_KEYBOARD_LL)
Quarantines h_{rogue} [DROP] while forwarding Operator KB [PASS]